

OLIST E-COMMERCE ANALYTICS

A Python and SQL analysis of a Brazilian online marketplace

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[1-ship-it/](#)

WHAT THIS PROJECT COVERS

Project overview

I worked with the Olist Brazilian e-commerce dataset: seven linked tables covering customers, orders, order items, payments, products, and sellers. My goal was to clean the raw data, join it into one usable table, and answer a structured set of business questions that a real analytics team would ask, from where customers are located to which sellers and categories actually drive revenue, and how loyal repeat buyers are.

7

source tables joined into one master table

15

business questions, split into three difficulty tiers

3

levels: basic, intermediate, advanced

From raw CSVs to answers

1

Clean

Fixed a mislabeled column, removed duplicate geolocation rows, and checked for orphan records and invalid prices before trusting the data.



2

Merge

Joined order items to orders, customers, products, and sellers into one master table. Kept payments separate, since they sit at order level, not item level.



3

Engineer

Added year and month fields, delivery timing, and order-level revenue so I could group and rank without repeating the same joins.



4

Analyze

Grouped, ranked, and tracked the data across three tiers, from simple counts to retention and year-over-year growth.

What I built it with

Python

Core language for cleaning, merging, and feature engineering

Pandas / NumPy

Grouping, joins, rolling averages, and correlation

Matplotlib

Ten charts covering trends, ranking, and concentration

Google Colab

Notebook environment, shareable and runnable end to end

SQL

A parallel version of the same 15 questions, written as queries

GitHub

Version control and public presentation of the finished project

TIER ONE

Basic questions

Get familiar with the data before drawing any conclusions from it

1

Unique customer cities

How wide the marketplace actually reaches

2

Orders placed in 2017

A baseline order count before comparing across years

3

Total sales per category

Which categories carry the business

4

Share of orders paid in installments

How much customers rely on financing to buy

5

Customers per state

Where demand concentrates geographically

TIER TWO

Intermediate questions

Looking for trends and patterns inside the sales and order data

1

Orders per month in 2018

Seasonality within a single year

2

Average items per order, by customer city

Whether some cities buy in bulk more than others

3

Revenue share by category

Concentration, not just raw totals

4

Price vs. purchase count correlation

Whether cheaper products really sell more often

5

Seller revenue, ranked

Who the top sellers are and how concentrated that group is

TIER THREE

Advanced questions

Turning the data into customer-level and strategic insight

1

Moving average of order value, per customer

Whether individual customers spend more or less over time

2

Cumulative monthly sales, per year

The pace of sales within each year

3

Year-over-year growth rate

Whether the business is actually growing

4

Six-month retention rate

Share of customers who buy again within six months of their first order

5

Top three spenders per year

The highest-value customers in each calendar year

WHY IT MATTERS

Each tier answers a different question

BASIC

Is the business set up the way I think it is?

Confirms geography, order volume, and category mix before anyone builds a strategy on top of assumptions.

INTERMEDIATE

Where should attention go this quarter?

Flags which cities, categories, and sellers matter most, so marketing and seller management aren't guessing.

ADVANCED

Is the business actually growing, and with whom?

Tracks growth rate, retention, and top customers, which is what leadership and finance actually ask for.

SKILLS DEMONSTRATED

What this project shows I can do

1

Cleaning and validating raw, multi-source data before analysis

2

Joining tables at the correct grain, without inflating row counts

3

Aggregating time-series data by month and by year

4

Testing a numeric relationship with correlation, not just a guess

5

Building a retention measure from raw order timestamps

6

Ranking and segmenting customers and sellers by value

WHAT'S NEXT

Where I'm taking this next

I want to run this notebook against the full dataset end to end, turn the strongest findings into a one-page dashboard, and check the Python results against the SQL version of the same fifteen questions to make sure both tell the same story.

Full notebook and source data

github.com/mdfarhan825301-ship-it

E-Commerce Public Dataset (customers, orders, order items, payments, products, sellers, geolocation)

Thank you

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